Great Place To Work. Certified FEB 2023 – FEB 2024 INDIA

UPES

UNIVERSITY OF TOMORROW

nirf RANKING 2023

#RANKED 52 IN INDIA

#University Category

QS WORLD UNIVERSITY RANKINGS

NO.1 PVT. UNIVERSITY IN ACADEMIC REPUTATION IN INDIA

NAAC

ACCREDITED GRADE 'A' BY NAAC

PERFECT SCORE OF 150/150 AS A TESTAMENT TO EXECEPTIONAL E-LEARNING METHODS

UPES UNIVERSITY OF TOMORROW

Elements of AIML

Miscellaneous – Lecture 01: PCA, LDA, and ICA in One View

Tofik Ali

School of Computer Science, UPES Dehradun

April 18, 2026

Supplementary lecture: **what each method optimizes, when to use it, and how they differ**

Repository: `https://github.com/tali7c/Elements-of-AIML`

Sources: Jolliffe and Cadima, "Principal component analysis: a review and recent developments," *Philosophical Transactions of the Royal Society A*, 2016.;Fisher, "The use of multiple measurements in taxonomic problems," *Annals of Eugenics*, 1936.;Hyvärinen and Oja, "Independent component analysis: algorithms and applications," *Neural Networks*, 2000.;Bishop, *Pattern Recognition and Machine Learning*, selected sections on PCA, LDA, and ICA.;Scikit-learn documentation, accessed 2026-04-18.

Agenda

Context

Method Intuition

Mathematics

Detailed Comparison

Summary

Learning Outcomes

By the end of this lecture, you should be able to:

- explain the core objective of **PCA**, **LDA**, and **ICA**,

Learning Outcomes

By the end of this lecture, you should be able to:

- explain the core objective of **PCA**, **LDA**, and **ICA**,
- identify whether a method is supervised or unsupervised,

Learning Outcomes

By the end of this lecture, you should be able to:

- explain the core objective of **PCA**, **LDA**, and **ICA**,
- identify whether a method is supervised or unsupervised,
- compare the three methods across assumptions, outputs, and use-cases,

Learning Outcomes

By the end of this lecture, you should be able to:

- explain the core objective of **PCA**, **LDA**, and **ICA**,
- identify whether a method is supervised or unsupervised,
- compare the three methods across assumptions, outputs, and use-cases,
- choose a method more carefully for a new AIML problem.

Reading Order and Prerequisites

- First read **Unit 03 Lecture 03** for the PCA pipeline and the leakage-safe preprocessing rule.

Reading Order and Prerequisites

- First read **Unit 03 Lecture 03** for the PCA pipeline and the leakage-safe preprocessing rule.
- For **LDA**, basic classification language from **Unit 04** is useful because labels matter.

Reading Order and Prerequisites

- First read **Unit 03 Lecture 03** for the PCA pipeline and the leakage-safe preprocessing rule.
- For **LDA**, basic classification language from **Unit 04** is useful because labels matter.
- For the eigenvalue derivation itself, use **Miscellaneous Lecture 02**.

Three Questions Before Choosing a Method

1. Do I have class labels?

Three Questions Before Choosing a Method

1. Do I have class labels?
2. Do I want **maximum variance**, **class separation**, or **independent sources**?

Three Questions Before Choosing a Method

1. Do I have class labels?
2. Do I want **maximum variance**, **class separation**, or **independent sources**?
3. Is my main goal compression, classification support, visualization, or source separation?

PCA: What Does It Do?

Principal Component Analysis

Find orthogonal directions that capture the **largest variance** in the data.

- Unsupervised: uses only the feature matrix.

PCA: What Does It Do?

Principal Component Analysis

Find orthogonal directions that capture the **largest variance** in the data.

- Unsupervised: uses only the feature matrix.
- Produces new components that are linear combinations of the original features.

PCA: What Does It Do?

Principal Component Analysis

Find orthogonal directions that capture the **largest variance** in the data.

- Unsupervised: uses only the feature matrix.
- Produces new components that are linear combinations of the original features.
- Common goals: compression, denoising, and visualization.

LDA: What Does It Do?

Linear Discriminant Analysis

Find projections that make classes far apart while keeping samples of the same class close together.

- Supervised: requires class labels.

LDA: What Does It Do?

Linear Discriminant Analysis

Find projections that make classes far apart while keeping samples of the same class close together.

- Supervised: requires class labels.
- Uses between-class scatter and within-class scatter.

LDA: What Does It Do?

Linear Discriminant Analysis

Find projections that make classes far apart while keeping samples of the same class close together.

- Supervised: requires class labels.
- Uses between-class scatter and within-class scatter.
- Common goal: prepare features for classification.

ICA: What Does It Do?

Independent Component Analysis

Recover latent components that are as **statistically independent** as possible.

- Unsupervised, but not a variance-maximization method.

ICA: What Does It Do?

Independent Component Analysis

Recover latent components that are as **statistically independent** as possible.

- Unsupervised, but not a variance-maximization method.
- Often motivated by source separation: mixed signals $\rightarrow$ original sources.

ICA: What Does It Do?

Independent Component Analysis

Recover latent components that are as **statistically independent** as possible.

- Unsupervised, but not a variance-maximization method.
- Often motivated by source separation: mixed signals $\rightarrow$ original sources.
- Common goals: artifact removal, blind source separation, latent-signal recovery.

Core Objective Functions

- **PCA:** maximize variance of projections.

$$\max_{\|v\|=1} v^\top C v$$

Core Objective Functions

- **PCA:** maximize variance of projections.

$$\max_{\|v\|=1} v^\top C v$$

- **LDA:** maximize class separation relative to class spread.

$$J(W) = \frac{|W^\top S_B W|}{|W^\top S_W W|}$$

Core Objective Functions

- **PCA:** maximize variance of projections.

$$\max_{\|v\|=1} v^\top C v$$

- **LDA:** maximize class separation relative to class spread.

$$J(W) = \frac{|W^\top S_B W|}{|W^\top S_W W|}$$

- **ICA:** maximize independence, usually through non-Gaussianity after centering and whitening.

What Matrix Problem Appears?

- **PCA:** eigenvectors of the covariance matrix C.

This is a key difference: PCA and LDA are directly tied to eigen-analysis; ICA goes beyond second-order statistics.

What Matrix Problem Appears?

- **PCA:** eigenvectors of the covariance matrix C.
- **LDA:** generalized eigenvectors of $S_W^{-1} S_B$.

This is a key difference: PCA and LDA are directly tied to eigen-analysis; ICA goes beyond second-order statistics.

What Matrix Problem Appears?

- **PCA:** eigenvectors of the covariance matrix C.
- **LDA:** generalized eigenvectors of $S_W^{-1} S_B$.
- **ICA:** often starts with whitening, but the final optimization is not just a covariance-eigenvalue problem.

This is a key difference: PCA and LDA are directly tied to eigen-analysis; ICA goes beyond second-order statistics.

Dimensionality Limits

Method	Label use	Dimension limit
PCA	No	choose any $k \leq d$
LDA	Yes	at most $C - 1$ for C classes
ICA	No	usually same order as observed signals/features

Detailed Comparison I

Method	Supervised?	Optimizes	Best used for
PCA	No	maximum variance	compression, visualization, denoising
LDA	Yes	class separation	classification-oriented projection
ICA	No	independence	source separation, latent-signal extraction

Detailed Comparison II

Method	Important assumption	Component relation	Typical risk
PCA	high variance is informative	orthogonal	loses label information
LDA	class labels are meaningful	not just variance-based	weak when class overlap is high
ICA	latent sources are independent and often non-Gaussian	independent, not merely orthogonal	unstable if assumptions fail

What Students Often Confuse

- PCA is **not** a classification method.

What Students Often Confuse

- PCA is **not** a classification method.
- LDA is **not** just "supervised PCA"; the optimization target is different.

What Students Often Confuse

- PCA is **not** a classification method.
- LDA is **not** just "supervised PCA"; the optimization target is different.
- ICA is **not** the same as PCA with more components; it cares about independence, not variance.

What Students Often Confuse

- PCA is **not** a classification method.
- LDA is **not** just "supervised PCA"; the optimization target is different.
- ICA is **not** the same as PCA with more components; it cares about independence, not variance.
- Orthogonality and independence are different ideas.

Which Method Should I Choose?

- Need 2D visualization or compression before modeling? **Start with PCA.**

Which Method Should I Choose?

- Need 2D visualization or compression before modeling? **Start with PCA.**
- Need a low-dimensional space that separates known classes? **Consider LDA.**

Which Method Should I Choose?

- Need 2D visualization or compression before modeling? **Start with PCA.**
- Need a low-dimensional space that separates known classes? **Consider LDA.**
- Need to separate mixed signals or hidden sources? **Consider ICA.**

Which Method Should I Choose?

- Need 2D visualization or compression before modeling? **Start with PCA.**
- Need a low-dimensional space that separates known classes? **Consider LDA.**
- Need to separate mixed signals or hidden sources? **Consider ICA.**
- If labels are unavailable, LDA is not applicable.

Summary

- PCA maximizes variance and is best for compression or visualization.
- LDA uses labels and is best when class separation matters.
- ICA seeks independent latent sources and is common in signal-style problems.
- The right method depends on the **goal**, not just on the number of dimensions.

Sources: Jolliffe and Cadima, "Principal component analysis: a review and recent developments," *Philosophical Transactions of the Royal Society A*, 2016.;Fisher, "The use of multiple measurements in taxonomic problems," *Annals of Eugenics*, 1936.;Hyvärinen and Oja, "Independent component analysis: algorithms and applications," *Neural Networks*, 2000.;Bishop, *Pattern Recognition and Machine Learning*, selected sections on PCA, LDA, and ICA.;Scikit-learn documentation, accessed 2026-04-18.

Exit Questions

1. Why is PCA unsupervised while LDA is supervised?
2. Why can LDA return at most $C - 1$ components?
3. What is the difference between orthogonality and independence?
4. Give one realistic use-case for ICA.